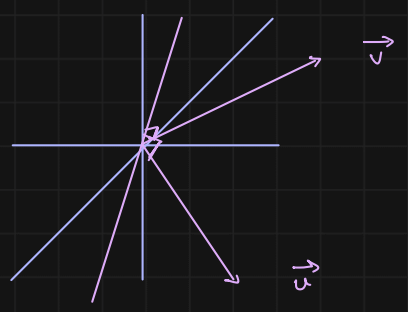
LECTURE 4

Proposition — <u>Cross Product in $\mathbb{R}^3$</u>

Given $\vec{u}, \vec{v} \in \mathbb{R}^3$, find a vector orthagonal to both.



A special sol'n is the <u>cross product</u>.

$$\vec{u} \times \vec{v} = \begin{bmatrix} u_2 v_3 - u_3 v_2 \\ u_3 v_1 - u_1 v_3 \\ u_1 v_2 - u_2 v_1 \end{bmatrix}$$

Proof — Compute $(\vec{u} \times \vec{v}) \cdot \vec{u} = 0$

$(\vec{u} \times \vec{v}) \cdot \vec{v} = 0$

Note that $\| \vec{u} \times \vec{v} \| = \| \vec{u} \| \, \| \vec{v} \| \sin \theta$

Order <u>does</u> matter.

Proposition — <u>Properties of cross products</u>

(1) $\vec{u} \times \vec{v} = -(\vec{v} \times \vec{u})$ anti-symmetry

(2) $\vec{u} \times (\vec{v} + \vec{w}) = \vec{u} \times \vec{v} + \vec{u} \times \vec{w}$

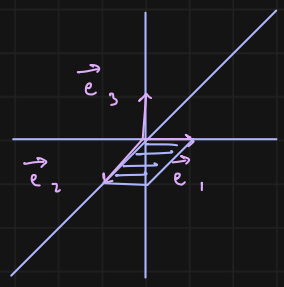
(3) $(c\vec{u}) \times \vec{v} = \vec{u} \times (c\vec{v}) = c(\vec{u} \times \vec{v})$

cross product, properties of cross product

CONTINUED

Ex

Let $\vec{e}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, $\vec{e}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$, $\vec{e}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$.



$$\begin{matrix} u_1 & u_2 & u_3 & u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 & v_1 & v_2 & v_3 \end{matrix}$$

$$\vec{e}_1 \times \vec{e}_2 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \times \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Note that $\vec{e}_2 \times \vec{e}_3 = -\vec{e}_3$

Remark

(1) The cross product is only defined for $\mathbb{R}^3$

(2) The cross product of vectors is a vector.

(3) Concepts carry from $\mathbb{R}^n$ to $\mathbb{C}^n$, except angles, dot product / length

Def'n

Let $\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_k \in \mathbb{F}^n$. The vector $\vec{v} = a_1 \vec{v}_1 + a_2 \vec{v}_2 + \ldots + a_k \vec{v}_k$ with $a_1, a_2, \ldots, a_k \in \mathbb{F}$ is a <u>linear combination</u> of $\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_k$.

Ex

In $\mathbb{F}^2$, $2 \begin{bmatrix} 1 \\ 3 \end{bmatrix} + (-1) \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \end{bmatrix}$

is a linear combination of $\begin{bmatrix} 1 \\ 3 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

Let $\vec{x}, \vec{y}, \vec{z} \in \mathbb{F}^n$. Then $\vec{x} = 1\vec{x} + 0\vec{y} + 0\vec{z}$ so $\vec{x}$ is a linear combination of $\vec{x}, \vec{y}, \vec{z}$.

$\vec{0} = 0\vec{x}_1 + 0\vec{x}_2 + \ldots + 0\vec{x}_k$. Then $\vec{0}$ is a linear combination of any combination of vectors.

linear combination

CONTINUED

Def'n Let $\vec{v}_1, \dots, \vec{v}_k \in \mathbb{F}^n$. Then the <u>span</u> of $\vec{v}_1, \dots, \vec{v}_k$ is

$$\text{Span}(\{\vec{v}_1, \dots, \vec{v}_k\}) = \text{set of all LC of } \vec{v}_1, \dots, \vec{v}_k$$
$$= \{ a_1 \vec{v}_1 + \dots + a_k \vec{v}_k : a_1, \dots, a_k \in \mathbb{F} \}$$
$$= \left\{ \sum_{i=1}^{k} a_i \vec{v}_i : a_i \in \mathbb{F} \right\}$$

Ex In $\mathbb{R}^2$, $\text{span}\left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\} = \left\{ a_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + a_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix} : a_1 \in \mathbb{R} \right\}$

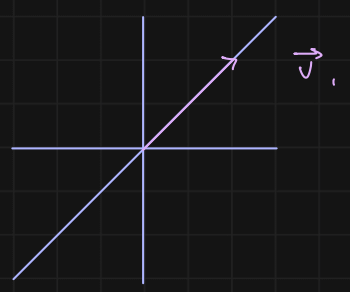
$$= \left\{ \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} : a_1, a_2 \in \mathbb{R} \right\} = \mathbb{R}^2$$

Is $\mathbb{R}^2$ $\text{Span}\left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ 4 \end{bmatrix} \right\}$? Yes.

<u>Geometry of spans in $\mathbb{R}^n$</u>

$\text{Span}(\{\vec{v}_1\}) = \{ a\vec{v}_1 : a \in \mathbb{R} \}$ = line through $(0, 0)$

In $\mathbb{R}^2$



$\text{Span}(\{\vec{v}_1, \vec{v}_2\})$, i.e. a plane

spans, geometry of spans in $\mathbb{R}^n$